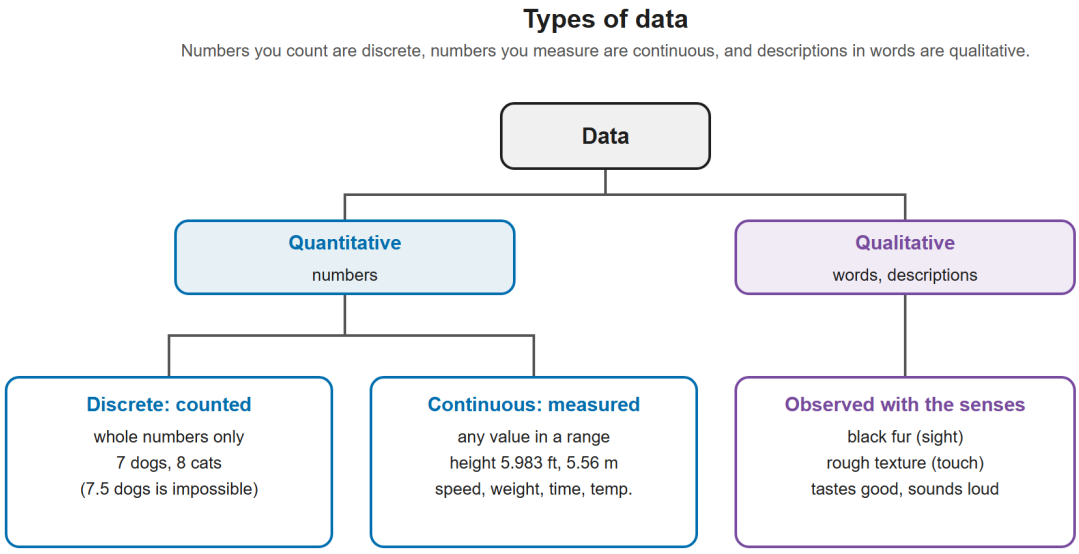
Lesson 1.3 - Quantitative vs Qualitative Data

> **Video:** Qualitative and Quantitative by The Organic Chemistry Tutor **Chapter:** 1 - Introduction to Statistics

Two types of data

```
                    Data
          ┌──────────┴──────────┐
    Quantitative              Qualitative
  (numbers / numerical)    (words / descriptive)
      ┌─────┴─────┐
  Discrete    Continuous
 (counted)    (measured)
```



The same tree as above, with an example in each box.

Quantitative data - numbers

Quantitative data is **numerical**, such as 5, 8 or 12. It comes in two forms.

Discrete - things you **count**

- Only whole-number values (0, 1, 2, 3, ...); nothing in between.
- Example: **8 cats**. You cannot have 8.5 cats or 2.37 cats.

Continuous - things you **measure**

- Any value in a range, including decimals.
- Example: a distance of 5.56 m or 7.283 m. It does not have to be a whole number.

- Other examples: **distance, speed, weight, height, time, temperature**.

> **Rule of thumb:** counted → discrete; measured → continuous. Both are quantitative.

Qualitative data - words

Qualitative data is **descriptive**. It comes from **observation** and is described with **words, not numbers**. It often involves the five senses:

Sense	Example
Sight	Color (the fur is black)
Touch	Texture (soft, hard, rough)
Taste	The food tastes good / bad
Hearing	The volume is low / high
Smell	It smells good / bad / of nothing

Quick quiz

Statement	Quantitative or qualitative?	Discrete or continuous?	Why
There are **7 dogs**	Quantitative	**Discrete**	You count dogs; 7.5 dogs is impossible
The cat's fur is **black**	**Qualitative**	n/a	Color, described with a word
Michael is **6 feet tall**	Quantitative	**Continuous**	Height is measured; it could be 5.983 ft
The cat's fur is **rough**	**Qualitative**	n/a	Texture, described with a word

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Key takeaways

- **Quantitative** = numbers. **Qualitative** = words or descriptions.
- Quantitative data is **discrete** (counted, whole numbers) or **continuous** (measured, any value in a range).
- Qualitative data usually describes what you see, feel, taste, hear or smell.